Topics in Statistics: MATH 598-782
Assignment 2
Due date: Friday, October 30, 2026 at 11:59pm
To be submitted on myCourses

Question 1. "seatpos" dataset in R records measurements taken from car drivers and relates them to seat position.

Car seat position depending driver size

Access the data in R:

```
> install.packages("faraway")
> library(faraway)
> data(seatpos)
> seatpos$hipcenter = -seatpos$hipcenter
(turning the response to positive values)
> head(seatpos)
  Age Weight HtShoes    Ht Seated  Arm Thigh  Leg hipcenter
1  46    180   187.2 184.9   95.2 36.1  45.3 41.3   206.300
2  31    175   167.5 165.5   83.8 32.9  36.5 35.9   178.210
3  23    100   153.6 152.2   82.9 26.0  36.6 31.0    71.673
4  19    185   190.3 187.4   97.3 37.4  44.1 41.0   257.720
5  23    159   178.0 174.1   93.9 29.5  40.1 36.9   173.230
6  47    170   178.7 177.0   92.4 36.0  43.2 37.4   185.150
```

Source: "Linear Models in R" by Julian Faraway, CRC Press, 2004.

Description:

Car drivers like to adjust the seat position for their own comfort. Car designers would find it helpful to know where different drivers will position the seat depending on their size and age. Researchers at the HuMoSim laboratory at the University of Michigan collected data on 38 drivers.

Format: `The dataset contains the following variables:`

`Age: Age in years`

`Weight: Weight in lbs`

`HtShoes: Height in shoes in cm`

`Ht: Height bare foot in cm`

`Seated: Seated height in cm`

`Arm: lower arm length in cm`

`Thigh: Thigh length in cm`

`Leg: Lower leg length in cm`

`hipcenter: horizontal distance of the midpoint of the hips from a fixed location in the car in mm.`

(a) Generate a heatmap displaying the pairwise correlations among the covariates (the first 8 variables in the above list).

You could use any of the following functions in `R`: Assuming that you have stored all the covariates in a matrix "X".

Method 1:

```
# Compute correlation matrix
> cor_mat <- cor(X)  # remove response if needed
# Basic heatmap
```

```
> heatmap(cor_mat)
```

Method 2:

```
> install.packages("corrplot")  # if not installed
> library(corrplot)
> cor_mat <- cor(X)
> corrplot(cor_mat, method = "color")
# or
> corrplot(cor_mat, method = "number")
```

Method 3:

```
> install.packages("pheatmap")
> library(pheatmap)
> cor_mat <- cor(X)
> pheatmap(cor_mat)
```

(b) Using `R`, fit a linear regression model to this data with `hipcenter` as the response variable. Present the results and discuss the significance of the covariates in the fitted model.

(c) Using the AIC, BIC, and Mallow's Cp, perform variable selection and report final selected submodel(s) based on these criteria. Comment on your findings!

(d) Fit a linear model using the ridge regression. Use 5-fold CV to select a data-dependent value of λ, call it $\hat{\lambda}$. Report the final ridge-based estimates of the regression parameters.

Question 2. Repeat parts (a), (b), and (d) of Question 1 for the `Hitters` dataset discussed in class.

```
> library(ISLR2)
> names(Hitters)
## Removing missing data
> Hitters<-na.omit(Hitters)
> dim(Hitters)
> X<-model.matrix(Salary~., Hitters)[, -1]
> y<-Hitters$Salary
```

Question 3. Consider the multiple linear regression model

$$y = \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_d x_d + \varepsilon.$$

Let $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)^\tau$, and assume that $\boldsymbol{X}_{n\times p}$ is an orthonormal design matrix, i.e. $\boldsymbol{X}^\top \boldsymbol{X}/n = \boldsymbol{I}_d$. Denote the vector $\widehat{\boldsymbol{\beta}} = (\hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_d)^\top$ as the least squares estimate of the parameter vector $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_d)^\top$, where

$$\hat{\beta}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij} y_i \quad , \quad j = 1, \ldots, d.$$

Suppose that some of the regression coefficients β_j are not significant in the model. We have decided to perform a variable selection task by solving the minimization problem

$$\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} (y_i - \beta_1 x_{i1} - \beta_2 x_{i2} - \ldots - \beta_d x_{id})^2 + \sum_{j=1}^{d} p_\lambda(\beta_j) \right\} \tag{1}$$

where $p_\lambda(\cdot)$ is a pre-specified non-negative penalty function, and $\lambda \geq 0$ is a penalty (or tuning) parameter. Denote $\widetilde{\boldsymbol{\beta}} = (\tilde{\beta}_1, \tilde{\beta}_2, \ldots, \tilde{\beta}_d)^\top$ as the solution to the minimization problem (1). Show that:

(a) for the LASSO penalty:

$$\tilde{\beta}_j = \text{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda)_+$$

where $b_+ = b$ if $b > 0$, otherwise $b_+ = 0$.

(b) for the SCAD penalty: for some $a > 2$,

$$\tilde{\beta}_j = \begin{cases} \text{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda)_+ & ; \text{ if } |\hat{\beta}_j| \le 2\lambda \\ \left(\frac{a-1}{a-2}\right)\text{sign}(\hat{\beta}_j)\big(|\hat{\beta}_j| - a\lambda/(a-1)\big)_+ & ; \text{ if } 2\lambda < |\hat{\beta}_j| \le a\lambda \\ \hat{\beta}_j & ; \text{ if } |\hat{\beta}_j| > a\lambda. \end{cases}$$

(c) for the MCP penalty: for some $\gamma > 1$,

$$\tilde{\beta}_j = \begin{cases} \left(\frac{\gamma}{\gamma-1}\right)\text{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda)_+ & ; \text{ if } |\hat{\beta}_j| \le \gamma\lambda \\ \hat{\beta}_j & ; \text{ if } |\hat{\beta}_j| > \gamma\lambda. \end{cases}$$

(d) for the ADALASSO penalty: for some pre-specified weights $\omega_1, \ldots, \omega_d$,

$$\tilde{\beta}_j = \text{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda\omega_j)_+$$

Note: Show all of your calculations.

Question 4. Consider the Prostate Cancer Data analyzed in **Question 1** of **Assignment 1**. The data is posted on the course website. Standardize each of the covariates x_1-x_8 to have mean zero and unit variance. Call the standardized covariates as x_1^*-x_8^*. The log of variable psa is the response variable y in our analysis: $y = \log(psa)$.

Consider the multiple linear regression model

$$y_i = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \ldots + \beta_8 x_8^* + \varepsilon_i \ , \ i = 1, 2, \ldots, 97.$$

(a) The intercept β_0 is estimated by $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$. Fit the above model to the data using the ridge regression with the tuning parameter λ over a reasonable grid of your choice. Plot the estimated regression coefficients $\hat{\beta}_1(\lambda), \hat{\beta}_2(\lambda), \ldots, \hat{\beta}_8(\lambda)$ as a function of λ. Place all eight plots in a single panel.

Choose an optimal value of λ using 5-fold CV.

(b) Repeat part (a) using the four penalties discussed in **Question 2** above. Report the final selected submodel(s) by the four penalties.

Choose an optimal value of λ for each method using BIC.

Question 5. Let Y be a continuous response variable and $\boldsymbol{x} = (1, x_1, x_2, x_3, x_4, x_5)^\top$ be a vector of explanatory variables. Assume that the conditional density function of Y given $\boldsymbol{x}$ is a mixture of two normal linear regression models given by

$$f(y; \boldsymbol{x}, \boldsymbol{\Psi}) = \pi\ \phi(y; \boldsymbol{x}^\top \boldsymbol{\beta}_1, \sigma_1^2) + (1-\pi)\ \phi(y; \boldsymbol{x}^\top \boldsymbol{\beta}_2, \sigma_2^2)$$

where $0 < \pi < 1$ is called the mixing probability (or proportion), and

$$\boldsymbol{\beta}_k = (\beta_{k0}, \beta_{k1}, \beta_{k2}, \beta_{k3}, \beta_{k4}, \beta_{k5})^\top \quad ; \quad k = 1, 2$$

are the vectors of regression coefficients (including intercepts) of the two components of the mixture model. The $\phi(y; \boldsymbol{x}^\top \boldsymbol{\beta}_k, \sigma_k^2)$ stands for the density function of the normal distribution with variance σ_k^2 and mean

$$\boldsymbol{x}^\top \boldsymbol{\beta}_k = \beta_{0k} + \beta_{k1} x_1 + \beta_{k2} x_2 + \beta_{k3} x_3 + \beta_{k4} x_4 + \beta_{k5} x_5$$

and the density function is given by

$$\phi(y; \boldsymbol{x}^\top \boldsymbol{\beta}_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left\{ -\frac{(y - \boldsymbol{x}^\top \boldsymbol{\beta}_k)^2}{2\sigma_k^2} \right\}, \ k = 1, 2.$$

The 15×1 dimensional vector of all parameters of the model is

$$\boldsymbol{\Psi} = (\pi, \boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \sigma_1^2, \sigma_2^2).$$

The objective is to estimate $\boldsymbol{\Psi}$ using the maximum likelihood estimator (MLE), denoted by

$$\hat{\boldsymbol{\Psi}} = \text{argmax}_{\boldsymbol{\Psi}} l_n(\boldsymbol{\Psi})$$

with the log-likelihood

$$l_n(\boldsymbol{\Psi}) = \sum_{i=1}^{n} \log f(y_i; \boldsymbol{x}_i, \boldsymbol{\Psi}) = \sum_{i=1}^{n} \log \left\{ \pi \ \phi(y_i; \boldsymbol{x}_i^\top \boldsymbol{\beta}_1, \sigma_1^2) + (1 - \pi) \ \phi(y_i; \boldsymbol{x}_i^\top \boldsymbol{\beta}_2, \sigma_2^2) \right\}.$$

As discussed in class, a closed-form expression for $\hat{\boldsymbol{\Psi}}$ is not available, and numerical optimization algorithms are typically employed to obtain an approximation to the MLE. Here, we use the EM algorithm to obtain such approximation to the MLE.

(a) Write down the E- and M-steps of the EM algorithm discussed in the class for the above model. Derive the closed forms of the updated estimates $\pi^{(m+1)}$, $\boldsymbol{\beta}_1^{(m+1)}$, $\boldsymbol{\beta}_2^{(m+1)}$, $\sigma_1^{2(m+1)}$ and $\sigma_2^{2(m+1)}$ of the parameters of the model.

(b) Consider the mixture model with parameters values: $\pi = 0.4, \sigma_1^2 = \sigma_2^2 = 1$ and

$$\boldsymbol{\beta}_1 = (-1, 1, 0, 2.5, 0, 0) \ , \ \boldsymbol{\beta}_2 = (1.5, -1, 2, 0, 0, 3).$$

Given a design matrix $\boldsymbol{X}_{200\times 6}$, we have generated 10 random samples of sizes $n = 200$ of y: $y_{s,1}, y_{s,2}, \ldots, y_{s,200}$, from the above model, where $s = 1, 2, \ldots, 10$. Note that all the generated samples share the same design matrix $\boldsymbol{X}_{200\times 6}$.

For each generated random sample $y_{s,1}, y_{s,2}, \ldots, y_{s,200}$, using the EM algorithm in **Part (a)** provide the estimate $\hat{\boldsymbol{\Psi}}_s$ of all the parameters (after the convergence of the EM), for $s = 1, 2, \ldots, 10$.

The generated data are stored in the files

$$\textbf{Assign2-Y.txt} : 10 \times 200 \quad , \quad \textbf{Assign2-cX.txt} : 200 \times 6$$

and are posted on course website. You can read the two data files in **R**.

(c) Apply the Lasso method to the above dataset and report the resulting parameter estimates (MPLE), i.e.

$$\tilde{\mathbf{\Psi}}_\lambda = \text{argmax}_{\mathbf{\Psi}} \left\{ l_n(\mathbf{\Psi}) - \lambda \sum_{k=1}^{2} \sum_{j=1}^{d} |\beta_{kj}| \right\}$$

For a specified grid of tuning parameter values λ, fit the model to each simulated sample and use the BIC to select an appropriate value of λ for each sample.